Solar Charge Controller



Category: **Solar, Power & Energy** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Charge controller

What it is used for

A solar charge controller regulates power from a photovoltaic array to charge a compatible battery safely and efficiently. Correct system matching, polarity checks, configuration, and monitoring help protect the battery and connected equipment from electrical faults and improper charging.

Main parts

- PV input terminals receive power from the solar panels.
- Battery terminals connect the controller to the battery bank.
- Load terminals supply controlled DC power where fitted.
- Charge regulation circuit controls battery charging.
- Display or indicator lights show charging status and faults.
- Voltage and current sensors monitor system conditions where fitted.
- Protection features help manage overvoltage, overcurrent, and reverse polarity where supported.
- Communication port connects compatible monitoring equipment where fitted.

Before you start

1. Only trained personnel should install or service the solar charge controller.
2. Read the controller manual and confirm compatibility with the PV array, battery chemistry, battery voltage, and system current.
3. Inspect the controller, terminals, wiring, connectors, and enclosure for damage or moisture.
4. Check the PV array open-circuit voltage and expected current against the controller's maximum ratings.
5. Identify the manufacturer's required connection and disconnection sequence before starting work.
6. Isolate PV input, battery, and loads according to the manual; remember that PV wiring can remain live in daylight.
7. Use suitable insulated tools and electrical PPE for the system voltage and work conditions.

How to use

1. Mount the controller in a dry, ventilated location with the clearances specified by the manufacturer.
2. Confirm that all power sources are isolated as required and verify terminal labels and polarity.
3. Connect the battery using the specified sequence, cable size, and protective devices.

4. Configure the battery type, charging voltage, charging current, and other settings according to battery manufacturer requirements.
5. Connect the PV array using the approved sequence and verify correct polarity before energising.
6. Connect DC loads to the designated load terminals only when supported and correctly rated.
7. Switch on the system according to the controller instructions and observe status indicators or display readings.
8. Check battery voltage, charging current, PV input, temperature, and any fault indicators.
9. Stop and isolate the system if abnormal heating, smoke, unexpected voltage, or a fault occurs.
10. After testing, secure terminal covers, route cables safely, and record settings and measured readings.

⚠ Safety

- Never exceed the controller's maximum PV voltage, input current, or battery current rating.
- Do not reverse battery or PV polarity; incorrect connections can damage equipment or create a fire hazard.
- PV modules can produce dangerous voltage in daylight; do not assume the array is de-energised.
- Never disconnect live PV connectors under load unless the equipment and procedure explicitly permit it.
- Use correctly rated fuses, circuit breakers, cables, and protective earth connections where required.
- Do not open or repair internal electronics unless qualified and authorised.

Care & storage

- Keep ventilation openings clear and remove dust from the enclosure using an approved method.
- Inspect terminals, cable insulation, protective devices, and mounting points regularly.
- Check fault logs and charging settings at the recommended service interval.
- Keep the controller dry and protect it from direct heat, moisture, and corrosive environments.
- Arrange qualified service if charging becomes irregular or a fault persists.

Common mistakes to avoid

- Connecting the PV array before the battery when the manual requires battery-first connection.
- Selecting the wrong battery chemistry or charging voltage.
- Ignoring PV voltage limits or polarity markings.
- Working on solar wiring in daylight without appropriate isolation and precautions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.